

CIRO + ReliefChain

System Workflow & Crisis Response Flow

Integrated Crisis Detection, Decision Optimization, and Aid Distribution Workflow

1. Situation Begins

- Heavy rainfall impacts the city
- Roads begin flooding
- Traffic congestion increases
- Citizens post flood reports online



2. CIRO Detects Signals

- Weather API detects extreme rainfall
- Social media flood reports spike
- Traffic systems detect abnormal congestion
- Field reports confirm flooding



3. Crisis Classification

- CIRO fuses all incoming signals
- Confidence score reaches critical threshold
- System classifies an Urban Flood event
- Severity marked as HIGH priority



4. Prediction & Optimization

- Predict affected population
- Estimate spread risk and duration
- Check shelter capacity
- Optimize ambulance and rescue routing
- Simulate best response strategy



5. Antigravity Trace System

- Logs WHY decisions were made
- Tracks confidence evolution
- Records resource allocation logic
- Creates transparent audit trail



6. ReliefChain Activation

- Citizens request aid through mobile app
- CNIC identity verification performed
- Vulnerability score calculated
- Aid Ledger prevents duplicate claims



7. NGO Response & Delivery

- NGOs receive verified requests
- Emergency teams dispatched
- QR and biometric verification performed
- Aid delivered in real time



8. Recovery & Feedback Loop

- Aid Ledger updated instantly
- Fraud prevention maintained
- Delivery status tracked
- System improves future response decisions

CIRO + ReliefChain transforms disaster response into a real-time, intelligent, transparent, and coordinated crisis orchestration system.